

Part	Area	Core Task	Weight
Part A	Machine Learning	Build an insurance claim prediction model	25%
Part B	Deep Learning	Rebuild and improve the ML solution using a neural network	25%
Part C	LangChain + RAG	Build a RAG Q&A system using only the M&A PDF	30%
Part D	Scenario-Based Assessment	Finance, Telecom & Healthcare business cases	20%
	Total		100%

PART A — MACHINE LEARNING

Insurance Claim Prediction

Business Context

An insurance company wants to develop a predictive system that can determine whether a customer/policy is likely to result in an insurance claim.

You have been provided with an insurance dataset containing customer and insurance-related information.

The dataset contains the following variables:

- age
- sex
- bmi
- children
- smoker
- region
- charges
- insuranceclaim

Your objective is to build and evaluate a **classification model** that predicts insuranceclaim.

Task 1. Dataset Understanding

Load the dataset using Pandas and perform an initial assessment.

You must:

1. Display the first 5 records.
2. Display the last 5 records.
3. Determine the number of rows and columns.
4. Display all column names.

5. Display the data types.
6. Identify missing values.

Questions to Answer

1. Which variables are numerical?
2. Which variables are categorical?
3. What is the target variable?
4. Is the target variable balanced?

Task 2. Exploratory Data Analysis

Perform appropriate exploratory data analysis.

Create suitable visualizations for:

Univariate Analysis

- Age
- BMI
- Charges
- Insurance claims

Task 3. Data Preprocessing

Prepare the dataset for machine learning.

You must:

1. Separate independent variables and the target variable.
2. Encode categorical variables appropriately.
3. Split the dataset into training and testing sets.
4. Apply feature scaling where appropriate.
5. Ensure that preprocessing does not result in data leakage.

Question

Explain why preprocessing transformations should be learned from the training dataset rather than the complete dataset.

Task 4. Build Machine Learning Models

Build a **model**.

For model:

1. Train the model.
2. Generate predictions.
3. Generate prediction probabilities where applicable.
4. Evaluate the model.

Task 5. Model Evaluation

Evaluate model using:

- Accuracy

PART B — DEEP LEARNING

Improving the Insurance Prediction Model

You have now developed traditional machine learning models.

Your next task is to determine whether a **Deep Learning model** can provide a better solution.

Task 7. Design a Neural Network

Using **TensorFlow/Keras**, develop a feed-forward neural network for the same classification problem.

Your architecture should contain:

- Input layer
- One or more hidden layers
- Appropriate activation functions
- Output layer
- Appropriate output activation for binary classification

You must document:

- Number of hidden layers
- Number of neurons
- Activation functions
- Optimizer
- Number of epochs

Question

Explain why you selected the architecture.

Task 8. Train the Deep Learning Model

Train your neural network using the prepared training data.

You should:

1. Train the model.

Task 9. Overfitting and Model Improvement

Analyze the training curves.

Determine whether your model is:

- Underfitting
- Properly fitted
- Overfitting

Question

If training accuracy continues increasing while validation accuracy begins to decline, what does this indicate?

Task 10. Evaluate the Deep Learning Model

Evaluate your final neural network using:

- Accuracy

PART C — LANGCHAIN + RAG

M&A Knowledge Assistant

Business Context

An organization has an **M&A Playbook** containing information about:

- M&A transaction lifecycle
- NDA and confidentiality
- Due diligence
- Valuation
- Representations and warranties
- Indemnification
- Risk management

- Integration planning
- TargetCo case study
- Post-merger integration

The document itself states that it can serve as a knowledge base for a **Retrieval-Augmented Q&A system**. The playbook covers the major stages from initial discussions and NDA through due diligence, valuation, definitive agreement, closing and integration.

You must use ONLY the provided M&A PDF as the knowledge source.

Do **not** use:

- Internet search
- Wikipedia
- Other M&A PDFs
- External websites
- External documents
- Additional knowledge bases

Your RAG system must answer questions based on the **provided M&A Playbook only**.

Task 12. Load and Process the PDF

Using an appropriate LangChain PDF loader:

1. Load the M&A PDF.
2. Extract the document contents.
3. Divide the document into chunks.
4. Experiment with chunk size and overlap.
5. Report:

Questions

1. Why is document chunking required?
2. What happens if chunks are too small?
3. What happens if chunks are too large?
4. How did you select your chunk size?

Task 13. Generate Embeddings

Generate embeddings for the document chunks.

You may use an appropriate embedding model available to you.

Explain:

1. What is an embedding?
2. Why are embeddings required for RAG?
3. How does semantic similarity work in the retrieval process?

Task 14. Build a Vector Store

Store the document embeddings using a vector store such as:

- FAISS

The vector store must contain **only information from the provided M&A PDF**.

Task 15. Build the RAG Pipeline

Implement a LangChain-based RAG pipeline.

Your implementation should:

1. Accept a user question.
2. Retrieve relevant chunks.
3. Pass the retrieved context to the LLM.
4. Generate an answer.
5. Display the answer.
6. Display the relevant retrieved context/source information.

Task 16. RAG Prompt Engineering

Your RAG prompt must instruct the LLM to:

- Use only the retrieved context.
- Avoid hallucinating.
- Clearly state when information is unavailable.
- Provide a concise but complete answer.

For example, the system should behave like:

"If the answer cannot be found in the provided M&A Playbook, state that the information is not available in the document rather than generating an answer from general knowledge."

PART D — SCENARIO-BASED ASSESSMENT

This section tests your ability to determine **when to use ML, DL, GenAI or RAG**.

You are not required to build models for these scenarios.

Your answers should focus on:

- Problem formulation
- Technology selection
- Architecture
- Model selection
- Evaluation
- Business implications

Scenario 1 — Finance: Credit Default Prediction

A bank has 2 million customer records containing:

- Income
- Credit score
- Loan amount
- Employment history
- Previous defaults
- Transaction behavior
- Debt-to-income ratio

The bank wants to predict whether a new borrower will default.

Questions

1. Would you use ML or DL?
2. Which algorithms would you consider?
3. What should the target variable be?
4. Which evaluation metrics would you prioritize?
5. Why might recall be more important than accuracy?
6. How would you handle class imbalance?
7. Under what circumstances would you consider Deep Learning?

Scenario 2 — Finance: Financial Document Assistant

A financial institution has thousands of:

- Annual reports
- Financial statements
- Policy documents
- Regulatory documents
- Investment reports

An analyst wants to ask:

"What are the major risks mentioned in the company's latest annual report?"

Questions

1. Is this primarily an ML, DL or GenAI problem?
2. Would you use RAG or fine-tuning?
3. Why?
4. Design the architecture.
5. How would you reduce hallucination?
6. How would you evaluate retrieval quality?

Scenario 3 — Telecom: Customer Churn

A telecom company wants to predict customers likely to leave within the next 30 days.

Available data includes:

- Call duration
- Data consumption
- Complaints
- Recharge frequency
- Plan type
- Tenure
- Network quality
- Customer service interactions

Questions

1. What type of ML problem is this?
2. What would be the target variable?
3. Which models would you consider?
4. Which evaluation metric would you prioritize?
5. How would you convert model predictions into business actions?
6. How could GenAI complement the churn model?

Scenario 4 — Telecom: Network Operations Copilot

A telecom company maintains:

- Network incident logs
- Troubleshooting manuals

- SOPs
- Equipment documentation
- Engineer notes

An engineer asks:

"Cell site XYZ has experienced repeated packet-loss incidents. What troubleshooting steps should I follow?"

Questions

1. Why would RAG be appropriate?
2. What information should be indexed?
3. Design the RAG architecture.
4. How would you prevent the LLM from inventing troubleshooting procedures?
5. What metadata could improve retrieval?
6. What would happen if the relevant procedure does not exist in the knowledge base?

Scenario 5 — Healthcare: Patient Readmission

A hospital wants to predict whether a patient will be readmitted within 30 days.

Available data includes:

- Age
- Diagnosis
- Length of stay
- Previous admissions
- Medication count
- Lab results
- Comorbidities

Questions

1. Would you use ML or DL?
2. What would the target variable be?
3. Which evaluation metrics would you consider?
4. Why could false negatives be particularly important?
5. How would you explain the prediction to a healthcare professional?
6. What risks should be considered before deploying the model?

Scenario 6 — Healthcare: Clinical Knowledge Assistant

A healthcare organization wants an AI assistant that can answer questions using only its approved:

- Clinical guidelines
- Hospital SOPs
- Treatment protocols
- Policy documents
- Procedure manuals

The organization states:

"The assistant must not answer using information outside our approved documents."

Questions

1. Design the architecture.
2. Why is RAG appropriate?
3. What would the retrieval pipeline look like?
4. How would you handle questions for which no answer exists in the knowledge base?
5. How would you evaluate hallucination?
6. What safeguards would you implement?
7. Would you fine-tune the LLM? Explain your reasoning.